(3)

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

A	B	C	(B∪C)	A∩(B∪C)	(A∩B)	(A∩C)
1	1	1	1	1	1	1
1	1	0	1	1	1	0
1	0	1	1	1	0	1
1	0	0	0	0	0	0
0	1	1	1	0	0	0
0	1	0	1	0	0	0
0	0	1	1	0	0	0
0	0	0	0	0	0	0

(A∩B)∪(A∩C)
1
1
1
0
0
0
0
0

EQUAL (proved)

1

a) $(p \vee q) \vee r = p \vee (q \vee r)$

p	q	r	$(p \vee q)$	$(p \vee q) \vee r$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	T	T
F	F	T	F	T
F	F	F	F	F

EQUAL

$p \vee (q \vee r)$

p	q	r	$(q \vee r)$	$p \vee (q \vee r)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	T	T
F	T	F	T	T
F	F	T	T	T
F	F	F	F	F

$(p \vee q) \vee r = p \vee (q \vee r)$

- they are equal. (all T, the last one is F)

b) $(p \wedge q) \wedge z = p \wedge (q \wedge z)$

p	q	z	$(p \wedge q)$	$(p \wedge q) \wedge z$	$(q \wedge z)$	$p \wedge (q \wedge z)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	F	T	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

$(p \wedge q) \wedge z = p \wedge (q \wedge z)$

• they are equal. (All False, the first one is T

(2)

p	q	z	$p \to q$	$q \to z$	$(p \to q) \wedge (q \to z)$	$p \to z$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	F	T
T	F	F	F	T	F	F
F	T	T	T	T	T	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

FINAL

FINAL: T T T T T T T T

All is True, that means it is tautology.